CH 1 A SQUARE AND A CUBE – PRACTICE QUESTIONS

INTRODUCTION	
1)	Write the first five square numbers.
2)	The next two numbers in the number pattern 1, 4, 9, 16, 25 ... are (a)35, 48 (b) 36, 49 (c) 36, 48 (d) 35,49
3)	Which of these options correctly classifies the numbers as perfect and non-perfect square numbers?

	Perfect Square numbers	Non perfect square numbers
option (a)	615,972,1396	576, 784, 961, 1024, 1225
option (b)	576, 784, 1024, 1225	615, 961, 972, 1396
option (c)	576, 784, 961, 1024, 1225	615, 972, 1396
option (d)	615, 972, 1024, 1396	576, 784, 961, 1225

PROPERTIES OF PERFECT SQUARES **1) ENDING DIGITS OF A NUMBER AND ITS SQUARE**	
4)	Which of the following numbers are perfect squares? (i) 484 (ii) 625 (iii) 576 (iv) 941 (v)961 (vi) 2500 (vii) 189 (viii) 225 (ix) 2048 (x) 343 (xi) 441 (x) 2916
5)	What will be the unit's digit of the squares of the following numbers? (i) 71 (ii) 599 (iii) 2783
6)	The following numbers are not perfect squares. Why ? 1057, 23453, 7928, 22342, 64000, 505050
7)	Which of the following end with digit 1? $123^2, 77^2, 82^2, 109^2$
8)	Write 5 numbers for which you cannot decide whether they are squares or not.
9)	The digit in the unit's place in the square root of 15876 is (a) 8 (b)6

	(c)4 (d)2
10)	A number ends in the digit 1. What could be the possible unit digit of the square root of that number? (a)1 or 9 (b) 3 or 6 (c) 6 or 9 (d) 7 or 9
11)	The digit at unit's place of a square of a number is 2 more than the digit at unit place of that number. Which of these can be the digit at the unit's place of the square of the number? (a) 1 (b) 3 (c) 5 (d) 9
12)	Find all two-digit numbers ending in 5 whose square ends with 25.
13)	How many zeros are there at the end of the square of a number ending in 10^n?
14)	Can a number ending in 45 be a perfect square?
15)	A student claims that numbers ending with 4 are always perfect squares. Is this true?
16)	The last digit of a number is 0, but the digit before is 3. Can it be a square number?
17)	A number ends in 01. Can it be a perfect square?
18)	A student finds a number ending with 4 and says it is a perfect square. Give an example where this is false.
19)	What is the least square number ending with exactly four zeros?
20)	A number ends with three zeros and is multiplied by 10. Can the result be a perfect square?
21)	A number ends with eight zeros. How many zeros will its square root end with?

PROPERTIES OF PERFECT SQUARES

2) PERFECT SQUARES AS SUM OF CONSECUTIVE ODD NUMBERS STARTING FROM 1

22)	Express 49 as the sum of 7 odd natural numbers
23)	Without adding, find the sum. (iii) 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23
24)	(i) Express 144 as the sum of some odd numbers. (ii) Express 324 as the sum of some odd numbers.
25)	Which of these can be expressed as the sum of first 8 odd natural numbers? (a) 4^2 (b) 8^2 (c) 9^2 (d) 16^2
26)	Consider the expression below. 1 + 3 + 5 + 7⋯⋯.+(2k + 1), where k is a natural number Which of these options describes the given expression? (a) The sum of the given expression is $(k + 1)0$. (b) The sum of the given expression is $(k - 1)0$. (c) The sum of the given expression is $k0$. (d) The sum of the given expression is $(2k + 1)0$.
27)	Show that sum of first 15 odd numbers = 225 = 15^2.
28)	A student claims the sum of the first 20 odd numbers is 420. Is this correct? Justify using the property of square numbers.
29)	Without squaring, show that $1 + 3 + 5 + \cdots + 37 = 19^2$.
30)	A student says $1 + 3 + 5 + \cdots + 47 = 576$.

	Is he correct?
31)	Find how many odd numbers must be added to get 625.
32)	If the sum of first k odd numbers is 784, find k.
33)	A boy adds the first 17 odd numbers to compute an area. What is the area of the square?
34)	If the sum of the first x odd numbers equals the perimeter of a square garden of side 26 m, find x.
PROPERTIES OF PERFECT SQUARES **3)PARITY OF PERFECT SQUARES**	
35)	Determine whether the squares of the following numbers are even or odd? (i) 213 (ii) 3824 (iii) 9777 (iv) 40028
36)	Which of the following is the square of an odd number? (a) 256 (b) 361 (c) 144 (d) 400
PROPERTIES OF PERFECT SQUARES **4) NUMBER OF NON-PERFECT SQUARES BETWEEN TWO PERFECT SQUARES**	
37)	How many natural numbers lie between squares of the following natural numbers? (i) 12 and 13 (ii) 25 and 26 (iii) 89 and 100
38)	How many non perfect square numbers lie between the squares of 15 and 16 ? (a) 31 (b) 32 (c) 30 (d) 35
39)	Find the number of non-square numbers lying between 40^2 and 41^2. a) 80

	b)40 c)30 d)84
40)	Consider the three consecutive numbers *a*, *b* and *c*. If there are 54 non-perfect square numbers between *a* and *b*, how many non-perfect square numbers lie between the numbers *b* and *c*? (a) 54 (b) 56 (c) 58 (d) 59
PROPERTIES OF PERFECT SQUARES **4) ODD PERFECT SQUARES CAN BE WRITTEN AS SUM OF TWO CONSECUTIVE NATURAL NUMBERS**	
41)	Express 81 as the sum of two consecutive numbers.
42)	What is the difference between 4 squares of any two consecutive numbers? Justify your answer.
43)	Find two consecutive perfect squares whose difference is 39.
44)	Two consecutive odd square numbers differ by 80. Find the numbers.
PROPERTY OF SQUARES – NO. OF FACTORS OF PERFECT SQUARES	
45)	Which number has an odd number of factors: 49 or 35? Why?
46)	A number has exactly 11 factors. Can it be a perfect square? Explain with reason.
PYTHAGOREAN TRIPLETS	
47)	Write a Pythagorean triplet whose one member is (i) 6 (ii) 18 (iii) 14 (iv) 16
48)	Which of the following triplet is a Pythagorean Triplet: (a) (4,5,6)

	(b) (11,60,61) (c) (10,8,6) (d) both b and c
49)	One of the members of the Pythagorean Triplet is given below. Find the other numbers. (a) (6,__,__) (b)(9,__ ,__) (c) (__,8,__) (d) (__, 24,__)
50)	Write a Pythagorean triplet if the smallest number is 9.
51)	Write two Pythagorean triplets each having one of the numbers as 5.
52)	A 5.5m long ladder is leaned against a wall. The ladder reaches the wall to a height of 4.4m. Find the distance between the wall and the foot of the ladder.
53)	The product of the numbers 14 and 16 is determined by using $(p + 1)(p - 1) = p^2 - 1$, where p is one of the members of the Pythagorean triplet which is neither the largest nor the smallest. Which of the following represents all the members of the Pythagorean triplet? (a) 4, 15 and 17 (b) 8, 15 and 17 (c) 14, 15 and 16 (d) 8, 63 and 64
54)	The hypotenuse of a right triangle with its legs of lengths 3x × 4x is (a) 5x (b) 7x (c) 16x (d) 25x

PATTERNS OF PERFECT SQUARES

55) Observe the following pattern and find the missing digits.

$11^2 = 1\ 2\ 1$

$101^2 = 10201$

$1001^2 = 1002001$

$100001^2 = 1\ \ldots\ 2\ \ldots 1$

$10000001^2 = \ldots$

56) Observe the following pattern and supply the missing number.

$11^2 = 1\ 2\ 1$

$101^2 = 1\ 0\ 2\ 0\ 1$

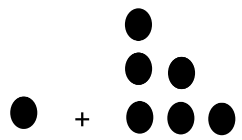
$10101^2 = 102030201$

$1010101^2 = \ldots\ldots\ldots\ldots.$

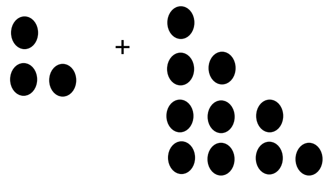
$\ldots\ldots\ldots\ldots.^2 = 10203040504030201$

57) Which of these will give a square number?

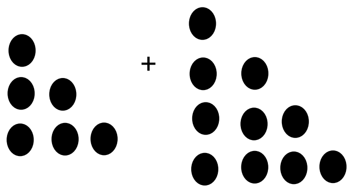
(a)



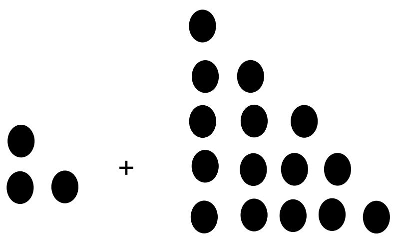
(b)



(c)



(d)



58)	A boy thinks 8400 can be arranged in a square array of dots. Is he correct?
FINDING THE SQUARE OF A NUMBER WITHOUT MULTIPLICATION	
59)	Find the square of the following numbers without actual multiplication: (a) 19 (b) 23 (c) 45 (d) 52
60)	How many factors does a square of a prime number have?
61)	A perfect square is multiplied by 4. Will the result also be a perfect square? Justify.
62)	Find the smallest number which is a perfect square and has exactly 3 different prime factors.
63)	The square of a number lies between 2025 and 2209. Find the number.
FINDING SQUARE ROOTS 1) **<u>Finding square root through repeated subtraction</u>**	
64)	Find the square root of 121 by repeated subtraction method
65)	Some steps to find the square root of a number are shown below. • 49 – 1 = 48 • 48 – 3 = 45 • 45 – 5 = 40 • 40 – 7 = 33 • 33 – 9 = 24 • 24 – 11 = 13 • 13 – 13 = 0 What is the square root of the number?
66)	The steps to determine the square root of a number m are shown below. $m - 1 = m - 1$ $m - 1 - 3 = m - 4$ $m - 4 - 5 = m - 9$ $m - 9 - 7 = m - 16$ ….. ……

	$m - 81 - 19 = 0$ Which of these options shows the value of m and its square root? **(a)** $m = 25$ and $\sqrt{m} = 5$ **(b)** $m = 81$ and $\sqrt{m} = 9$ **(c)** $m = 100$ and $\sqrt{m} = 10$ **(d)** $m = 121$ and $\sqrt{m} = 11$
FINDING SQUARE ROOTS **<u>Finding the square root through prime factorisation :</u>**	
67)	Show that each of the following numbers is a perfect square. Also find the number whose square is the given number in each case. (i) 1156 (ii) 2025 (iii) 14641 (iv) 4761
68)	Is 225 a perfect square? If so, find the number whose square is 225
69)	Show that 63504 is a perfect square. Also find the number whose square is 63504
70)	Show that 17640 is not a perfect square
71)	Find the smallest number by which given numbers must be multiplied so that the product is a perfect square. Also find the number whose square is the new number. (i) 180 (ii) 23805 (iii) 12150 (iv) 7688 (v) 8820 (vi) 3675 (vii) 605 (viii) 2880 (ix) 4056 (x) 3468
72)	Find the smallest number by which following numbers should be divided so that the result is a perfect square. Also find the number whose square is the new number (i) 14283 (ii) 1800 (iii) 2904 (iv) 16562 (v) 3698 (vi) 5103 (vii) 3174 (v) 1575
73)	Find the least number by which 500 must be multiplied to make it a perfect square (a) 2 (b) 3

	(c) 4 (d) 5
74)	Find the least perfect square number which is divisible by each of the numbers 4, 5 and 10.
75)	Find the smallest number by which 1250 must be multiplied to make it a perfect square. Also find the square root of the number so obtained.
76)	Is 2352 a perfect square? If not, find the smallest number by which 2352 must be multiplied so that the product is a perfect square. Find the square root of the quotient.
77)	Find the square root of 14641 by Prime Factorisation method.
78)	If x is the number of prime factors in the prime factorisation of 800, find $2x^2$
79)	Which of these is a prime factorisation of a square number? **(a)** 2 × 3 × 5 × 5 **(b)** 3 × 3 × 4 × 4 × 5 × 5 **(c)** 2 × 2 × 3 × 5 × 5 × 7 × 7 **(d)** 2 × 2 × 3 × 3 × 5 × 5 × 11 × 11
80)	The prime factorisation of a number is as shown. 2 × 2 × 3 × 3 × 3 × 3 What is the square root of the number? (a) 6 (b) 8 (c) 18 (d) 22
81)	A number d is such that it's prime factorisation consists of 12 prime numbers. Which of these shows the prime factorisation of the square root of the number? (a) 2 × 3 × 5 (b) 2 × 2 × 3 × 5 (c) 2 × 2 × 2 × 3 × 3 × 5 (d) 2 × 2 × 2 × 2 × 2 × 2 × 3 × 3 × 3 × 3 × 5 × 5
82)	The prime factorization of an even square number is as shown.

	$3 \times 3 \times 5 \times 5 \times 7 \times 7 \times 7 \times p$ What is the smallest value of p? (a) 2 (b) 4 (c) 14 (d) 28
83)	A number n is written as $p^2q^4r^3$, where p, q and r are the prime factors. Which expression represents a perfect square number and what is the square root of the number? **(a)** $\boldsymbol{\frac{p^2q^4r^3}{r}}$ ***and*** $\boldsymbol{\sqrt{\frac{p^2q^4r^3}{r}} = pq^2r}$ (b) $\boldsymbol{\frac{p^2q^4r^3}{p}}$ ***and*** $\boldsymbol{\sqrt{\frac{p^2q^4r^3}{p}} = pq^2r^2}$ **(c)** $\boldsymbol{p^2q^4r^3 \times p}$ ***and*** $\boldsymbol{\sqrt{p^2q^4r^3 \times p} = p^2\, q^2r}$ **(d)** $\boldsymbol{p^2q^4r^3 \times q}$ ***and*** $\boldsymbol{\sqrt{p^2q^4r^3 \times q} = pq^2r}$
84)	Find the smallest number that is a square & divisible by 50
85)	Find a perfect square number less than 100 and divisible by 2 and 3.
86)	Find smallest perfect square divisible by 18 and 50.
87)	Find the least number that must be multiplied to 750 so that it becomes a perfect square
88)	A number expressed as a product of primes is: $2^3 \times 3 \times 5^2 \times 7^4$. Find the least number to multiply to make it a perfect square.
FINDING SQUARE ROOTS **<u>3) Finding the square root by long division method:</u>**	
89)	Find the square root of the following numbers by division method: **[CW]** (a) 54756 (b) 12996
90)	Find the least number which must be subtracted from 7230 to make it a perfect square. Find the square root of the number so obtained.
91)	What least number must be added to 7344 to make it a perfect square?
92)	Find the least number which should be subtracted from 180 to make it a perfect square.

93)	Find the greatest number of two digits which is a perfect square.
94)	Find the least number of three digits which is a perfect square.
95)	Find the least number of four digits which is a perfect square.
96)	Find the greatest 4 digit number which is a perfect square.
97)	Find the greatest number of five digits, which is a perfect square.
98)	Find the smallest number of 6-digit which is a perfect square.
NUMBER OF DIGITS IN THE SQUARE ROOTS	
99)	The number of digits in the square root of 33856 is (a) 5 (b) 4 (c) 3 (d) 2
100)	A perfect square number having n digits where n is even will have square root with (a) $n+1$ digits (b) $\frac{n}{2}$ digits (c) $\frac{n}{3}$ digits (d) $\frac{n+1}{2}$ digits
101)	A perfect square number having n digits where n is odd will have square root with (a) $n+1$ digits (b) $\frac{n}{2}$ digits (c) $\frac{n}{3}$ digits (d) $\frac{n+1}{2}$ digits
SQUARE ROOT OF DECIMALS	

102)	Find the square root of the following decimal numbers: (a) 16.81 (b) 151.29 (c) 53.59 (d) 156.25 (e) 104.04 (f) 92.16
103)	The square root of $(0.09 + 2 \times 0.21 + 0.49)$ is (a) $\sqrt{0.09} + \sqrt{0.49}$ (b) $2\sqrt{0.21}$ (c) 1 (d)0.58
104)	Given that $\sqrt{4096} = 64$, the value of $\sqrt{4096} + \sqrt{40.96}$ is (a) 74 (b) 60.4 (c) 64.4 (d) 70.4
105)	Find the square root of $6\frac{1}{3}$ up to three places of decimals
106)	Find the square root of 90 by estimation method.
107)	A decimal number is multiplied by itself. If the product is 51.84, find the number.
108)	An integer k is obtained by multiplying a number c by itself. Consider the two statements about the square root of k. Statement I: The square root of k is c if $c > 0$. Statement II: The square root of k is $-c$ if $c < 0$. Which of these statements is correct? (a) Only Statement I (b) Only Statement II (c) Both Statements I and II (d) Neither Statement I nor II
WORD SUMS	
109)	8649 students are sitting in a lecture room in such a manner that there are as many students in a row as there are rows in the lecture room. How many students are there in each row?
110)	There are certain numbers of rows of trees in a garden. The number of trees in each row is

	twice the number of rows. If the number of trees in the garden is 1250, find the number of rows in the garden.
111)	A gardener has 1500 plants. He wants to plant these in such a way that the number of rows and the number of columns remain same. Find the minimum number of plants he needs more for this.
112)	A yoga instructor wants to arrange maximum possible number of 6000 students in a ground so that the number of rows is same as the number of columns. How many rows will be there if 71 students were left out after the arrangement?
113)	Area of a square plot is 2304 m^2. Find the side of the square plot.
114)	A rectangle ABCD has AB = 12 cm and BC = 6cm.Find the length of its diagonals correct to 2 decimal places.
115)	In an auditorium, the number of rows is equal to the number of chairs in each row. If the capacity of the auditorium is 2025, find the number of chairs in each row.
116)	The area of square field is 49284 m^2. Find the cost of fencing the field at Rs 15 per metre.
117)	In a school, all children were made to stand in square formation. Out of 3050 children,25 were left after forming the square. How many children are there in each row?
118)	The Cost of levelling a square lawn at rupees 15 per square meter is 19,935. Find the cost of fencing the lawn at rupees 22 per meter.
119)	Each member of a picnic party contributed twice as many rupees as the total number of members and the total collection was rupees 3042. Find the number of members present in the party.
120)	The area of a square field is 60025 m^2. A man cycles along its boundary at 18km/hr. In how much time will he return at the starting point?
121)	The length of a rectangle is 3 times its breadth. Its area is 972 sq. meter. Find the perimeter of the rectangle.
122)	Sarah is arranging square tiles to make a larger square pattern. She has 64 tiles total. Can she arrange all her tiles into a perfect square? If so, how many tiles will be on each side?
123)	A school playground has a square sandbox. The area of the sandbox is 144 square feet. What is the length of each side of the sandbox?

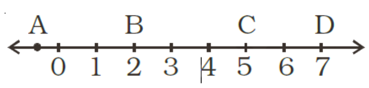
124)	Marcus is building a square garden bed. He wants the area to be exactly 81 square meters. How many meters of fencing will he need to go around the entire perimeter?
125)	A chess board has 64 squares arranged in a perfect square pattern. How many squares are along each edge of the chess board?
126)	A square pizza is cut into 100 equal square pieces arranged in rows and columns. How many pieces are in each row?
127)	A square parking lot has an area of 196 square yards. A car is parked in the exact center. How far is the car from each edge of the parking lot?
128)	A square swimming pool has an area of 400 square meters. Pool tiles that are 1 meter × 1 meter are placed around the entire edge. How many tiles are needed for the perimeter?
129)	Maria has a square piece of fabric with an area of 256 square inches. She wants to cut it into a smaller square that is half the width. What will be the area of the smaller square?
130)	A square courtyard has an area of 784 square meters. Benches are placed every 4 meters around the perimeter. How many benches are there total?
131)	A square tile floor has 529 individual tiles arranged in a perfect square pattern. During renovation, workers need to replace all tiles along the outer edge. How many tiles need to be replaced?
132)	Alex is creating a square mosaic using 841 small square pieces. If arranged in a perfect square, how many pieces will be along each edge? If each small piece is 2 cm × 2 cm, what is the total area of the mosaic?
133)	A square city block has an area of 900 square meters. Street lamps are installed at each corner and every 10 meters along each side. How many street lamps are needed for the entire perimeter of the block?
134)	A square swimming pool has an area of 400 square meters. Pool tiles cost ₹ 5 per meter for the

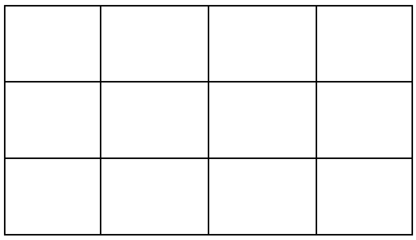
	perimeter. Calculate the total cost and justify your method.
135)	Compare two squares: one with area 144 square inches and another with perimeter 48 inches. Which has the larger area?
136)	Jake arranges 225 photos in a square pattern but wants to add one more row and column. How many additional photos does he need?
137)	A farmer has 100 meters of fencing to create a square field. What is the maximum area possible?
138)	Design a square courtyard with an area between 500-600 square meters. Determine the side length and explain why?
139)	Two squares have areas in the ratio 4:9. If the smaller square has a perimeter of 24 cm, find the side length of the larger square.
140)	The sum of Geetha's age and the square of her brother's age is 54. What is Geetha's age?
141)	A rectangular garden floor has an area of 225 m^2. Square tiles, each with a side length of 5 m, are used to cover the entire floor.
142)	A swimming pool deck has an area of 180 m^2. To cover the surface, workers use square anti-slip mats, each with a side length of 3 m.How many such square mats are needed to cover the entire deck?
143)	A rectangular plot has an area of 360 m^2. The farmer makes the biggest possible square inside it. Find the remaining area?
144)	A parking ground has a rectangular area of 289 m^2. The manager wants to use the largest possible square area inside it for VIP parking. What will be the area of the square ?
145)	A librarian arranged books in a perfect square with 26 books in each row. After removing some damaged books, she rearranged the remaining books into a smaller square with 2 fewer books per row. How many books did she remove?
146)	A farmer planted some trees when he counted ,he found that if he adds 15 more trees he will be

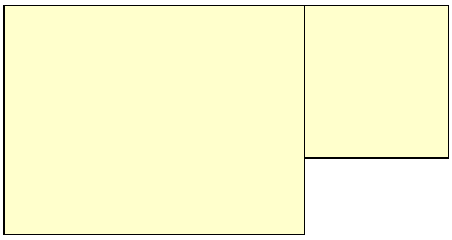
	able to arrange them in 21 rows and 21 columns ,How many trees did he have ?
147)	A school arranges chairs in a square pattern. When 5 chairs are removed, the arrangement is no longer a perfect square. What is the maximum possible number of chairs?
148)	A square picture frame has an area of 484 cm^2. If the border of width 2 cm is removed from all sides, what is the area of the remaining inner square?
149)	A school assembly forms a square of 961 students. If 31 more students join, can they still form a square?
150)	A rectangular field measures 24 m × 18 m. What is the side of the largest square plot that can fit inside?
151)	A square board of side 30 cm is cut into square tiles of side 5 cm. How many tiles are obtained?
152)	A square garden's area is equal to the sum of areas of four smaller squares whose sides are same natural numbers. If the side of the smaller square is 4 cm, find the side of the garden.
153)	If the perimeter of a square is numerically equal to its area, what is the side of the square?
154)	A square-shaped field is increased by 1 m on each side; its area increases by 45 m^2. Find the original side.
155)	A number is 45 more than a perfect square and 44 less than the next perfect square. What is the number?
156)	A classroom carpet covers 961 square units. If each tile is 1×1, how many tiles are along one side?
157)	A person walks around a square garden 4 times covering 240 m. What is the area of the garden?
158)	A school builds a square lawn. When the side is increased by 3 m, the area increases by 189 m^2. Find the original side length.
159)	A student wants to paint a square mural of area 1089 cm^2. If paint covers 25 cm^2 per brush dip, how many dips are required?

160)	A store wants to arrange 198 packets into a square stack. How many more packets are needed to complete the square?
161)	A village square park increases its boundary from 160 m to 200 m. How much does the area increase?
162)	A school arranges 784 chairs in a square. If 4 rows of chairs are removed, how many chairs remain?
163)	A square park has an area of 1296 m^2. A walking path of uniform width is built inside along all sides, reducing the area to 1156 m^2. Find the width of the path.
164)	A shopkeeper arranges 400 boxes in a square pile. If he removes 39 boxes, can the remaining boxes form a perfect square?
165)	A square field of area 625 m^2 is expanded by extending each side by 2 m. What is the new area?
166)	A square garden has area equal to the sum of the first 11 odd numbers. Find the side and area of the garden.
167)	Find the side of a square whose area is equal to the area of a rectangle with sides 6.4m and 2.5m.
168)	Rahul walks 12m north from his house and turns west to walk 35m to reach his friend's house. While returning, he walks diagonally from his friend's house to reach back to his house. What distance did he walk while returning?
169)	The perimeters of two squares are 40 and 96 metres respectively. Find the perimeter of another square equal in area to the sum of the first two squares.
EVALUATE	
170)	Area of a square plot is 729 m^2. The side of the square is: ' (a) 33m (b) 37m (c) 27m

	(d) 23m
171)	The value of $\sqrt{144} + \sqrt{1.44}$ is? (a) 24 (b) 13.2 (c) 1.32 (d) none of these
172)	The Simplified form of $\frac{3}{\sqrt{0.09}}$ is (a) 30 (b) 3 (c) 1 (d) none of these
173)	Evaluate $\sqrt{5^2 + 12^2}$ (a) 169 (b)13 (c) 289 d) none of these
174)	Find the value of $\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{169}}}}$
175)	Find the value of $\frac{\sqrt{289}}{\sqrt{361}}$
176)	Find the value of $\frac{\sqrt{3025}}{\sqrt{4225}}$
177)	Find the value of $\sqrt{45796}$ and hence find the quotient of $\sqrt{4.5796} \div \sqrt{457.96}$
178)	The number of digits in the square root of 33856 is (a) 5 (b) 4 (c) 3

	(d) 2
179)	Compare and order these values: $\sqrt{36}$, 7^2, and 45.
180)	Between which two consecutive integers does $\sqrt{75}$ lie?
181)	Solve for x: $x^2 = 196$
182)	$\frac{2707}{\sqrt{x}} = 27.07$, find x
183)	Find x: $3^4 + 3^2 = 9x$
184)	Find x: $175 \times x = 35^2$
185)	Find a square whose sum of digits is a perfect square.
186)	Find a square number less than 500 such that the number formed by reversing its digits is also a perfect square.
187)	A number leaves remainder 3 when divided by 4. Can it be a perfect square? Justify.
188)	A number is 5 less than a perfect square and 4 more than another perfect square. Find the number.
189)	Find two perfect squares whose sum is also a perfect square.
190)	Two consecutive numbers have the same number of factors. Prove that one of them cannot be a perfect square
191)	Without using calculators, determine if 1540 is closer to 39^2 or 40^2.
192)	Find the least perfect square greater than 921.
193)	Which letter best represents the location of $\sqrt{25}$ on a number line? 

	(a) A (b) B (c) C (d) D
194)	Find three numbers in the ratio 2:3:5, the sum of whose squares is 608.
CASE STUDY	
195)	For a school performance, an equal number of students are to be arranged in each row and column. (a) How many students can be added in a row/column in Arrangement 1 for 625 students? (b) The teachers organising the performance think that it will be difficult to arrange 625 students in one group. They think of dividing the group into two square arrangements. How many students can be there in each arrangement?
196)	A king wanted to reward his advisor, a wise man of the kingdom. So, he asked the wiseman to name his own reward. The wiseman thanked the king but said that he would ask only for some gold coins each day for a month. The coins were to be counted out in a pattern of one coin for the first day, 3 coins for the second day, 5 coins for the third day and so on for 30 days. Q.1 Without making calculations, find how many coins will the advisor get in that month?
197)	The rectangle below is made of unit squares. 1. What are the possible ways of converting it into a square? 2. How many more unit squares are needed to make it perfect square? 
198)	The area of two square playgrounds are 625 sq m and 256 sq m respectively. (a) Find the sides of each playground (b) Find the ratio of their sides

	(c) If both the squares are joined, find the perimeter of the new figure.
199)	In the given figure, find the length AB if areas of two squares are 81 sq cm and 25 sq cm. respectively. 
200)	A rectangular paper of length 45 cm and breadth 5 cm is cut into form a square with the same area. What is the side of the square?
201)	In the figure, side of bigger square is 19cm and smaller square is 17 cm. What is the area of the shaded portion? 
202)	A perfect square number has four digits, none of which is zero. The digits from left to right have values that are: even, even, odd, even. Find the number.